[Geophysical &] Radar Imaging

Andrea Monti Guarnieri DEIB – Bldg 21 (for the time being)

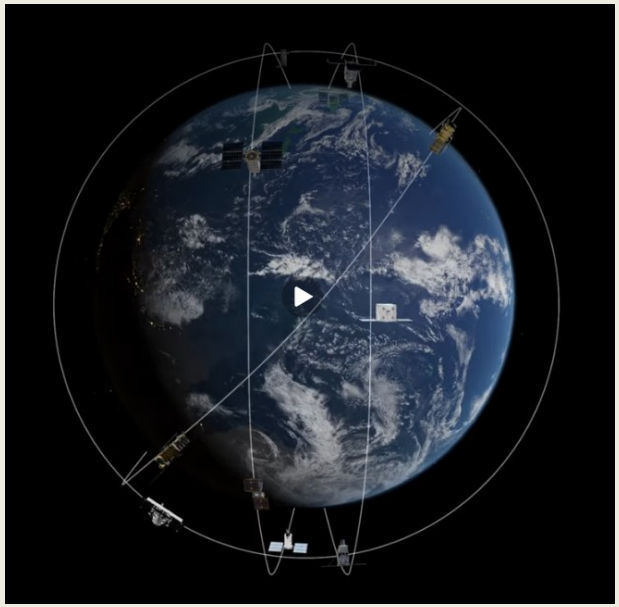
andrea.montiguarnieri@polimi.it

https://politecnicomilano.webex.com/join/andrea.montiguarnieri

What's new - 2026

Italy Brings IRIDE Earth Observation Service Online

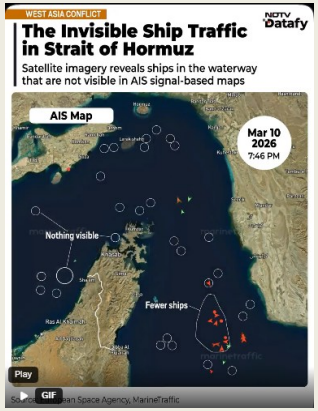
By Andrew Parsonson · July 6, 2026

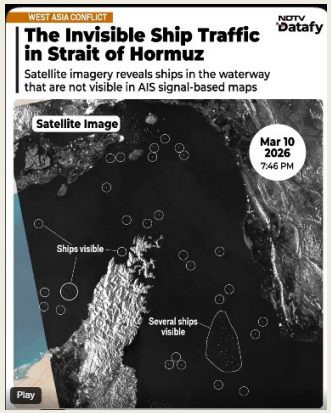




Sentinel-1D User Data Opening from 17 April 2026

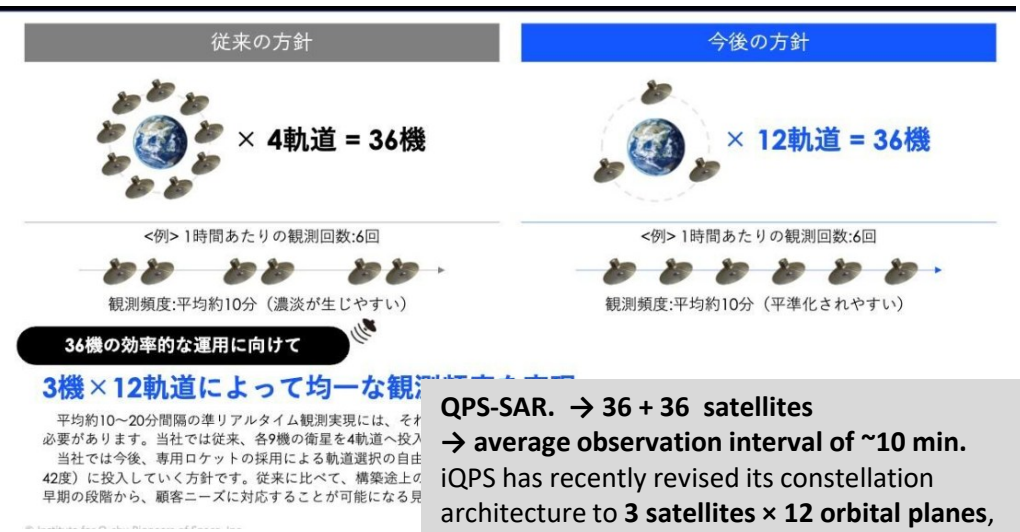
POLIMI SAR group participated to support, design and CalVAL of all ESA SAR, since 1991.



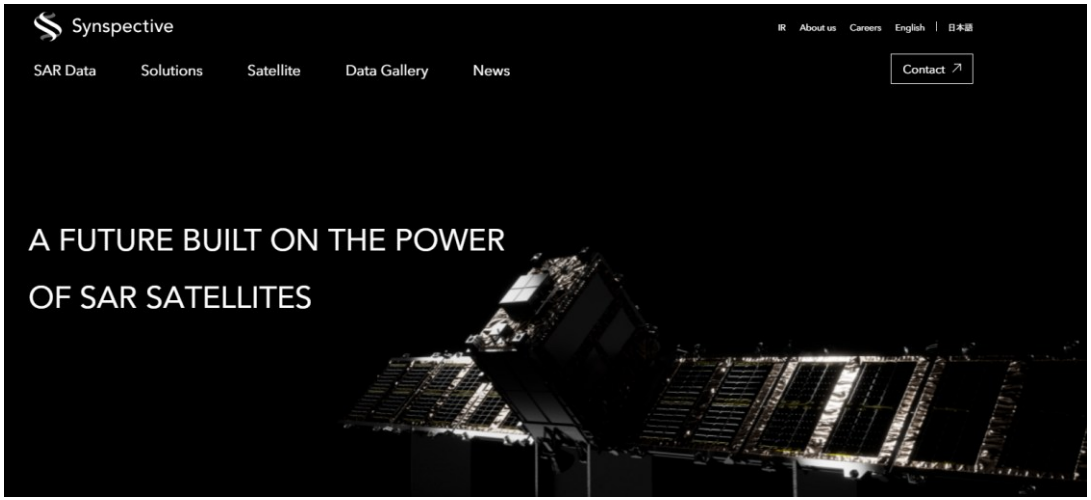


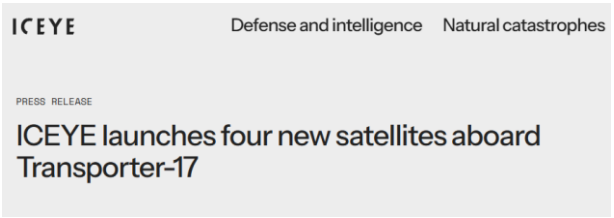
https://www.esa.int/Applications/Observing_the_Earth/IRIDE/IRIDE_satellites

Some of the new Radar (SAR) satellites launched in 2026

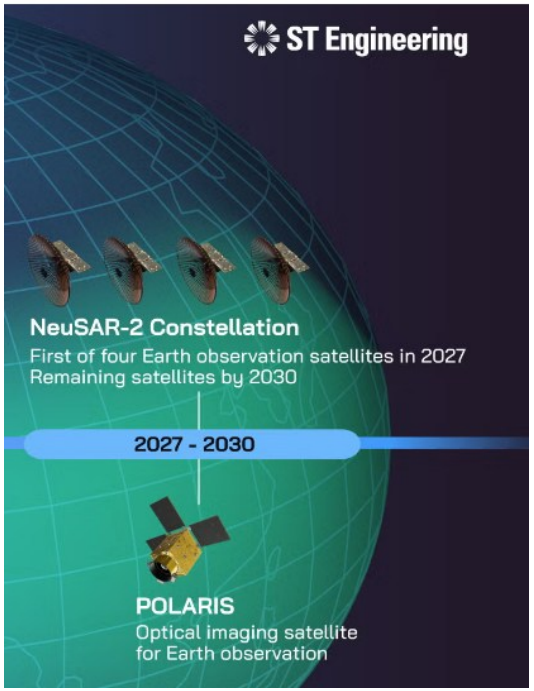


QPS-SAR. → 36 + 36 satellites
→ average observation interval of ~10 min.
iQPS has recently revised its constellation architecture to **3 satellites × 12 orbital planes**, with plans to further expand the constellation beyond the initial 36 satellites.





target of 100 annually by 2027 and beyond, supported by a matching launch cadence.



not only space SAR

Description | Table Of Content | Methodology | Download PDF

Automotive Radar Market by Range (Short Range, Medium Range, Long Range), Vehicle Type (PC, LCV, HCV), Frequency (2X-GHz and 7X-GHz), EV Type (BEV, PHEV, FCEV, HEV), Mounting (In-cabin, Exterior), Application, and Region - Global Forecast to 2033

CAGR 20.7% (2026-2033)	USD 27.03 BN MARKET SIZE, 2033	320 REPORT PAGES	280 MARKET TABLES

HEXAGON | IDS GeoRadar

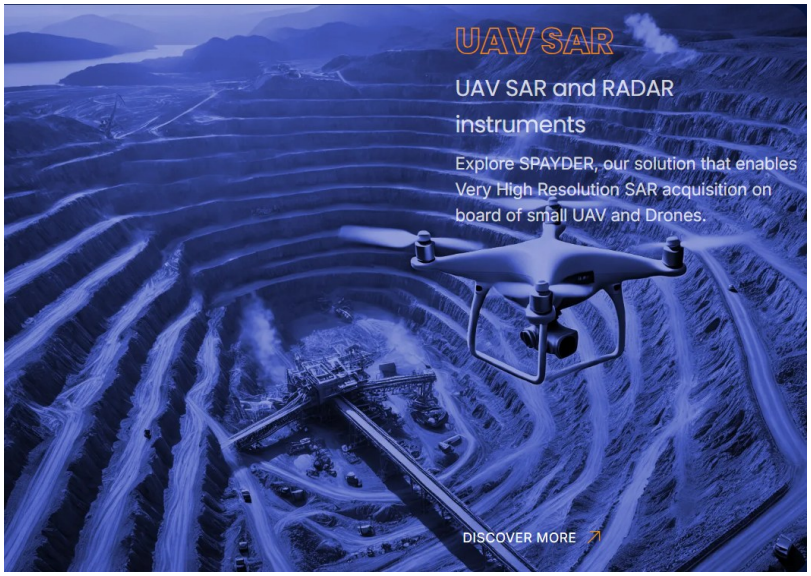
Industries Products Services About Us Contact Us Customer Area

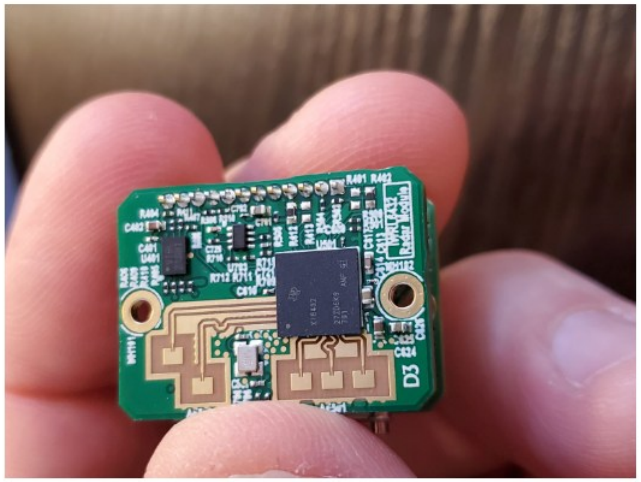
Home > About Us > News > IDS GeoRadar launches ArcSAR Neo

ArcSAR Neo

IDS GeoRadar, part of Hexagon, today announced the launch of ArcSAR Neo, the first fully 3D SAR for 360° mine slope monitoring







D3 Embedded mmWave Radar Sensor

Program

Remote sensing principles, methods and applications: from IR to visible. Review of principles: Plank law, radiance, reflectance & Kirchhoff law. Image generation, properties (resolution, field of view, noise, number of looks), and processing (calibration, geocoding, detection, data analysis and quality evaluation). Examples of applications of optical and radiometric images.

Coherent imaging. Review of plane wave, wave propagation, polarimetry. Speckle and its statistical properties, coherence and radiometric resolution. Example: Lidar. Antenna arrays and applications: change identification, high resolution location.

RADAR basics: Doppler Radar (identification of moving target), pulsed radars (precise location by ranging). Radar cross section, thermal noise and the Radar equation.
Image generation (back-projection)
Application: Imaging (Synthetic Aperture) Radars

RADAR systems and applications: Interferometric, tomographic, polarimetric and coherent change detection methods and examples.
Millimetric deformation estimation & MIMO systems for industrial, automotive, & drone applications.







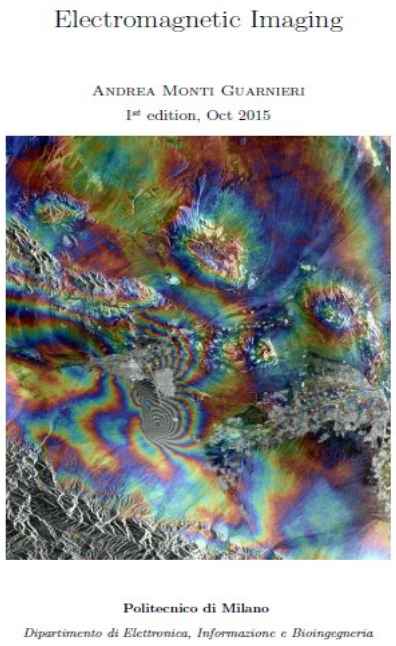
Text-books

1) Lessons slides, recording, and other material

2) Textbook: *Electromagnetic Imaging (also on webeep)*

200 pp: for your (future) profession,

< 25% required for the course



3) Complementary Textbook, to get deep into SAR Interferometry

InSAR Principles: Guidelines for SAR Interferometry Processing and Interpretation

three parts: dummy, user, expert



All material is freely available from webeep

Exams

Oral discussion – two options

1. **With home-class-works**
   Complete the two proposed homeworks during the course and at home.
   → The oral discussion starts from your work and results.
2. **Without home-class-works**
   → The oral discussion may start from any course topic: radar design, theory, or applications.

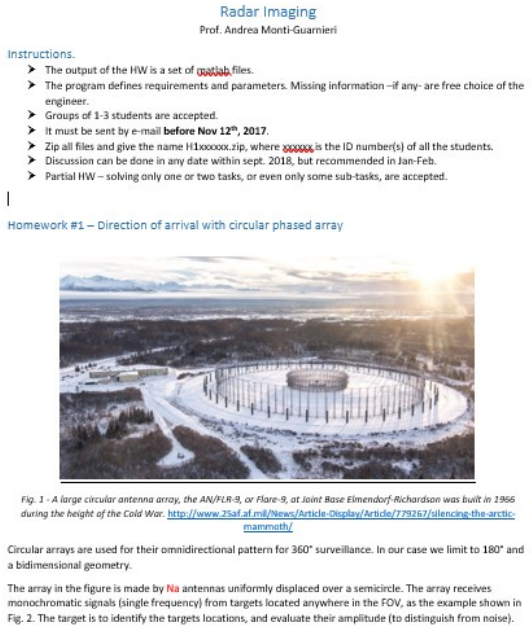
When ?

Any date - within the exam session – fix a date by e-mail

REMEMBER TO SUBSCRIBE TO THE EXAM

Home-class-work(s) in the AI era

The lab provides a unique opportunity to learn how to turn theory into a working system — a skill relevant to telecommunication systems in general, not only radar.



Radar Imaging

Prof. Andrea Monti-Guarnieri

Instructions.

- The output of the HW is a set of matlab files.
- The program defines requirements and parameters. Missing information –if any- are free choice of the engineer.
- Groups of 1-3 students are accepted.
- It must be sent by e-mail **before Nov 12th, 2017**.
- Zip all files and give the name H1xxxxxx.zip, where xxxxxx is the ID number(s) of all the students.
- Discussion can be done in any date within sept. 2018, but recommended in Jan-Feb.
- Partial HW – solving only one or two tasks, or even only some sub-tasks, are accepted.

Homework #1 – Direction of arrival with circular phased array

Fig. 1 - A large circular antenna array, the AN/FLR-9, or Flare-9, at Joint Base Elmendorf-Richardson was built in 1966 during the height of the Cold War. http://www.25af.af.mil/News/Article-Display/Article/779267/silencing-the-arctic-mammoth/

Circular arrays are used for their omnidirectional pattern for 360° surveillance. In our case we limit to 180° and a bidimensional geometry.

The array in the figure is made by Na antennas uniformly displaced over a semicircle. The array receives monochromatic signals (single frequency) from targets located anywhere in the FOV, as the example shown in Fig. 2. The target is to identify the targets locations, and evaluate their amplitude (to distinguish from noise).

- Two projects to be implemented in Matlab.
- Each with multiple questions of progressive difficulty.
- Covering the aspects discussed at lessons

- **Dedicated laboratory sessions** will support the projects: problem-solving methods, implementation, Q&A, and hands-on assistance.
- AI tools may be used to support the development of the projects.

Exam with projects - rules

- **Projects shall be completed within the deadline.** Submit a short report (Word, PowerPoint, …) summarizing:
  - your understanding of the problem and requirements;
  - how you address it through radar technology and system design
  - your formal/mathematical solution
  - how you validate your solution in MATLAB
  - your take-home message
- Projects shall be **discussed during the first exam session (January–March)**
- The report itself will **not be graded**.
- What will be assessed is your ability to **critically discuss your work, explain your choices, and answer questions** — in other words, **what you have actually understood**.

Project Report

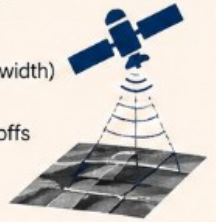
What to include (SAR imaging)

1 Problem and Requirements

- Context and objectives
- Scenario and data (if any)
- Performance requirements (e.g., resolution, coverage, accuracy)

2 Approach: Radar Technology & System Design

- System configuration (e.g., orbit, waveform, bandwidth)
- Processing chain
- Design choices and trade-offs

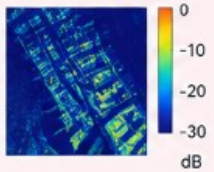


3 Formal / Mathematical Solution

- Signal and imaging model
- Key equations and algorithms $s(k_r) = \int \sigma(x)\, e^{-j2k_r R(x)}\, dx$
- Processing steps

4 Validation with MATLAB

- Implementation details
- Simulation / data used
- Results (images, plots, metrics)
- Comparison with requirements



5 Results and Take-Home Message

- Main results (images, metrics)
- Discussion
- Lessons learned
- Take-home message

Project Report

SAR Imaging Project

1. Problem and Requirements
2. Approach: Radar Technology & System Design



3. Formal / Mathematical Solution
4. Validation with MATLAB
5. Results and Take-Home Message

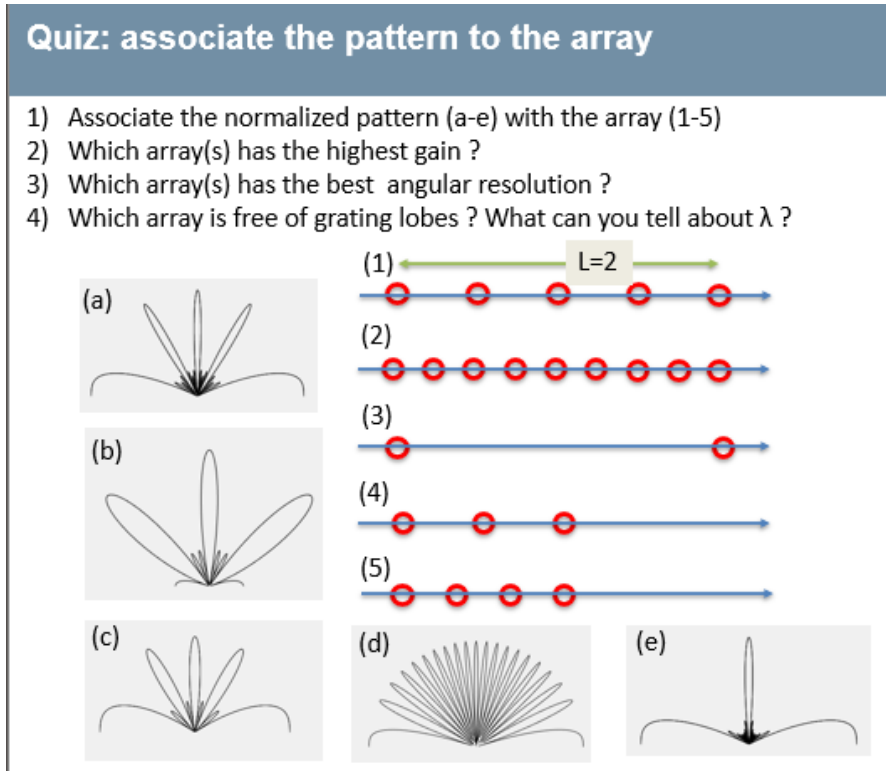
A clear and concise report helps you prepare the project discussion at the exam.

During lectures over the semester: Wooclap sessions

Wooclap sessions will be organized during lessons – usually at end of cycle

- mainly intended to check your understanding of the course (this is useful for you and for me)
- Optional, not required for the exam, however
- …. good performance may **contribute positively to the final grade:**

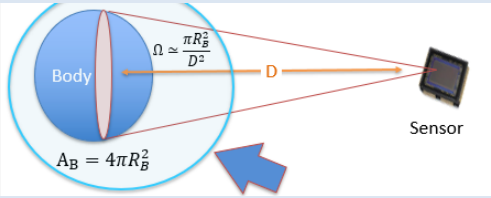
  you need to login to wooclap with your POLIMI account, for that.

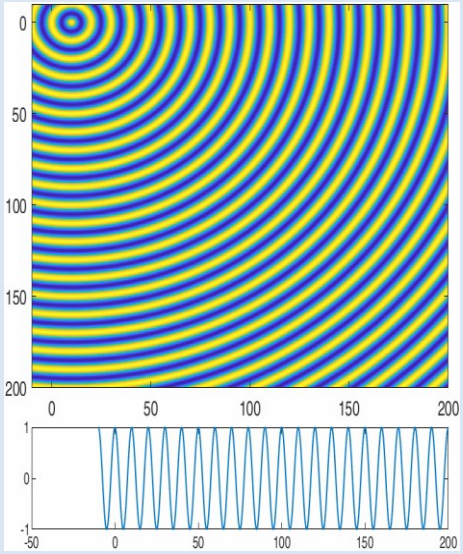


Course topics at a glance: part #1

Waves, antennas & optical/IR remote sensing

System (& maths)

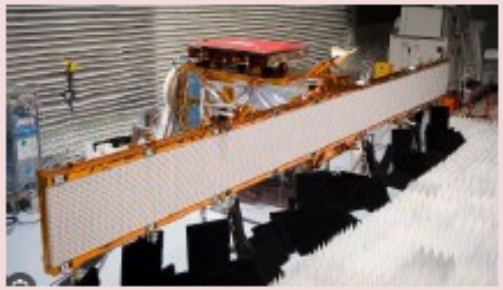




$$s_m(r, f_0, t) = \exp(j(2\pi f_0 t \pm 2\pi f_r r))$$

Technologies





Applications

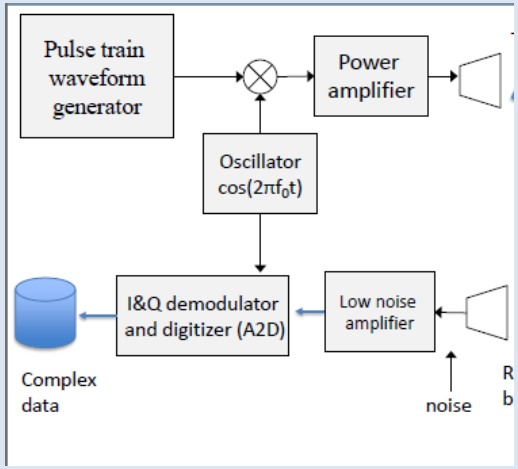


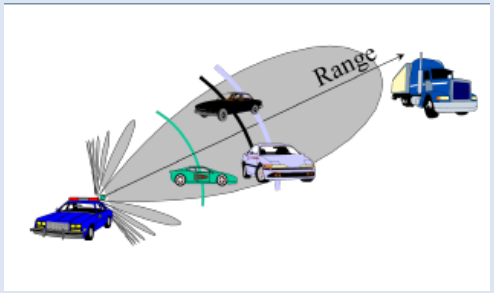


Course topics at a glance: part #2 and part #3

Radar principles, and Radar Image understanding

System (& maths)



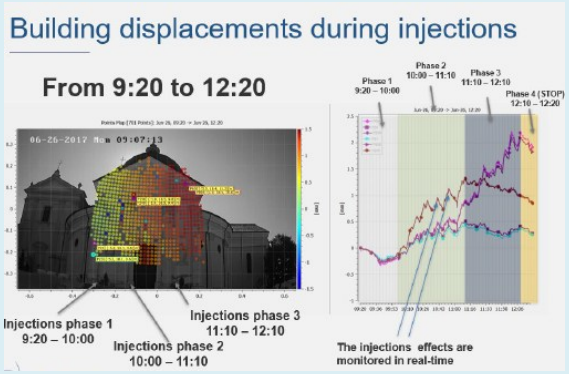


Technologies





Applications





What's after: job, research and industries

The Radar research group at POLIMI cooperates with:

universities & research all over the word

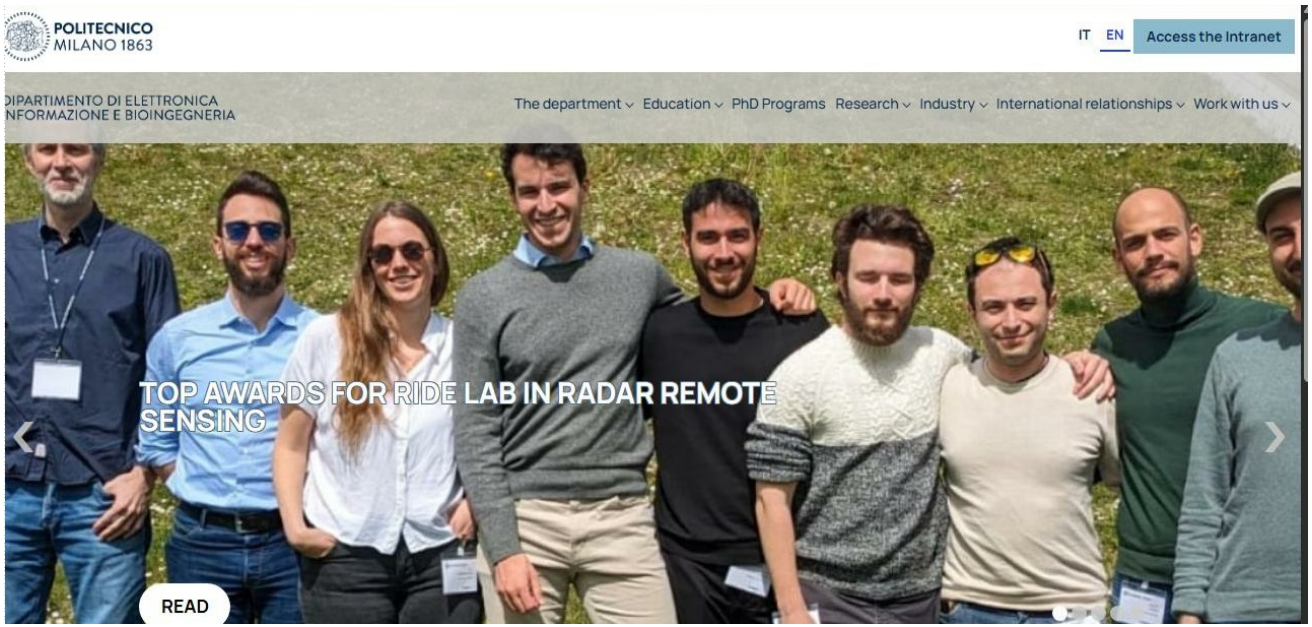
Industries – world wide with excellence in Italy (Rome and Milan / Lombardy)

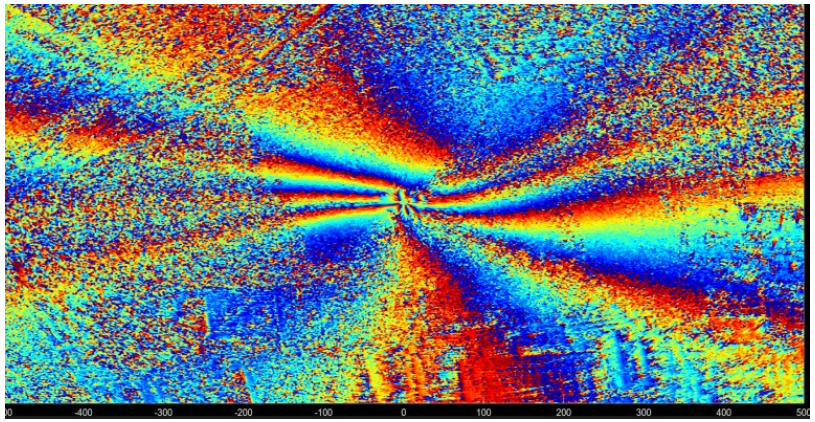
The course will open to job opportunities:

- **Industrial Radar application** (manufacturing, health monitoring, automotive, defence)
- **Remote sensing for environment** (civil protection, insurances, resources management)



Thesys and Ph.D tacks at RIDE-LABs POLIMI



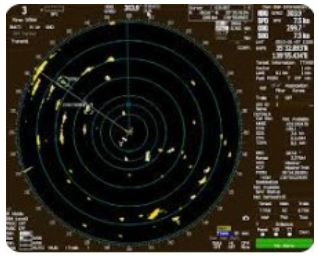




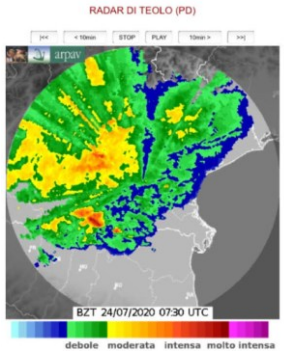
Introduction: Radar or SAR ?

Conventional Radars



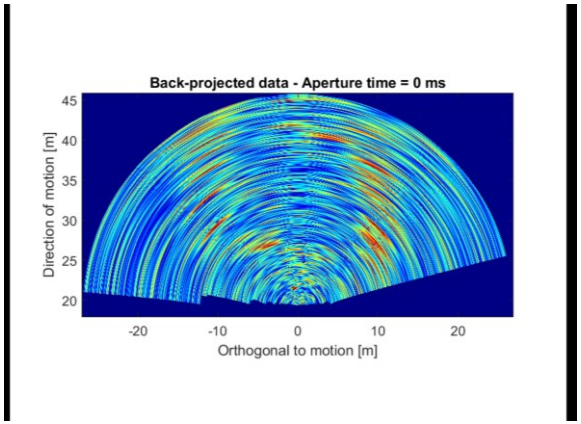






A SAR exploits the motion of the Radar platform to generate high-resolution images.
It is a synthetic array

Automotive-Radar->SAR





What makes Radar unique ?

The super-long wavelength opens a wide-variety of applications that cannot be achieved otherwise:

1. Penetration
2. Sensitivity to roughness
3. Precise & fast estimation of distance
4. Active (no need for light sources)
5. Propagation over the horizon
6. Sensitivity to moisture

The major drawback – need for huge antennas is solved by the synthetic aperture. This enables high resolution imaging from the space.

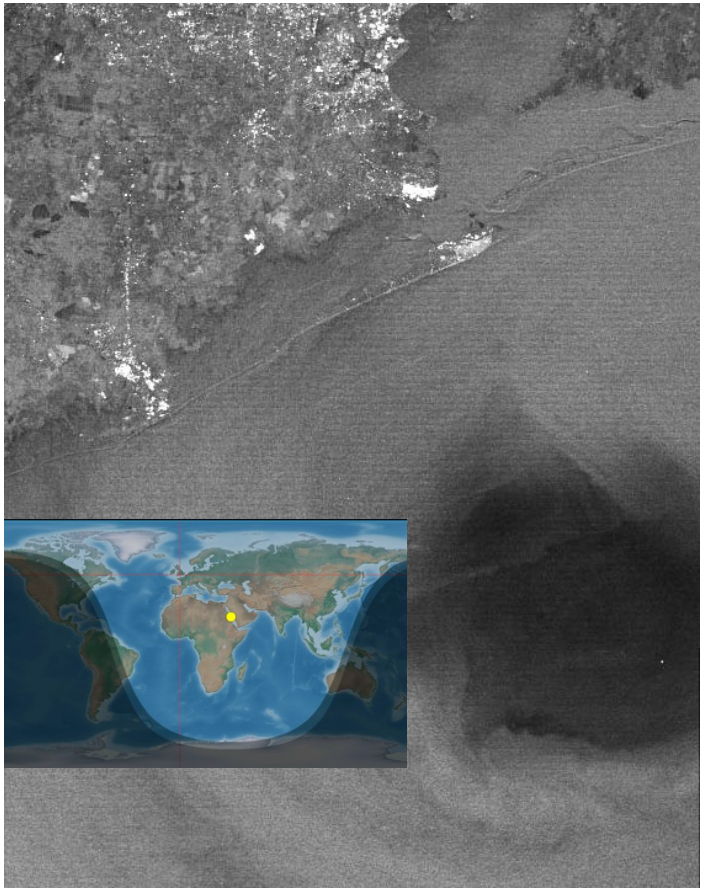
Capabilities, benefits and limitations will be covered by course.

Killer applications: in depth penetration

Optical



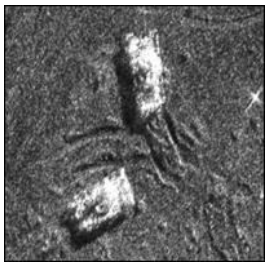
RADAR



Penetration: other of many examples



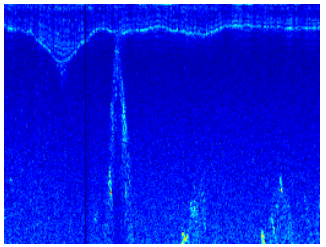


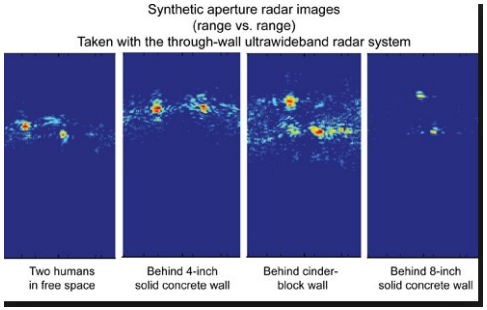


Thorugh the forest

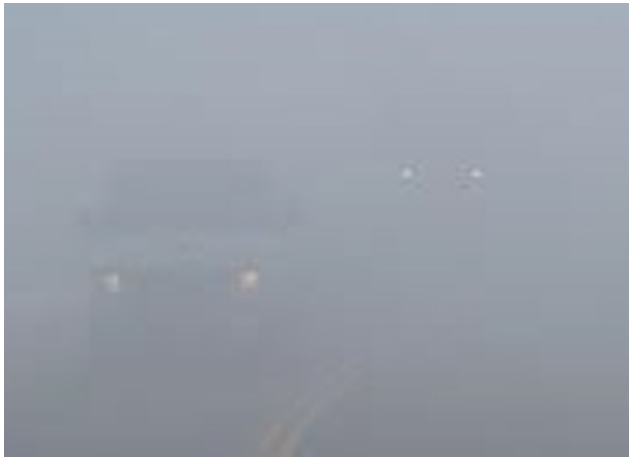


… a glacier





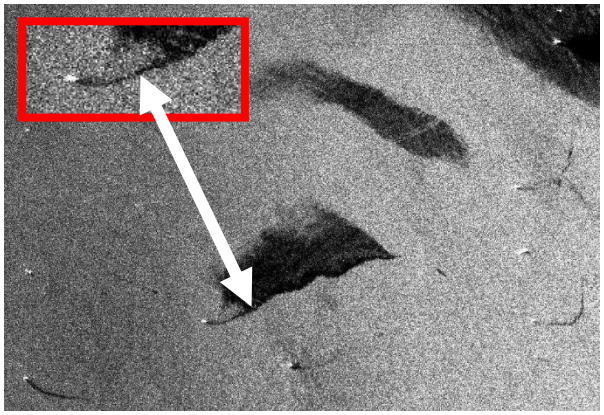
Through the wall



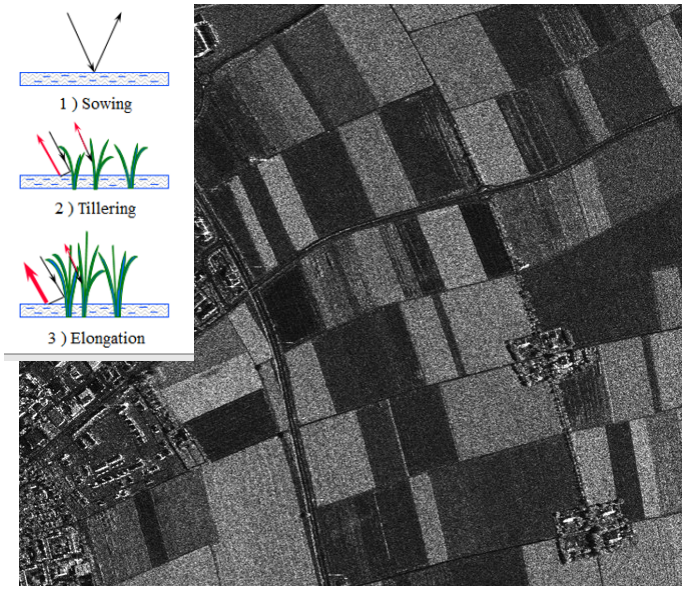
Driving in the fog

Each application needs proper selection of wavelength, Radar configuration, observable (amplitude, phase, coherence) and calibration.

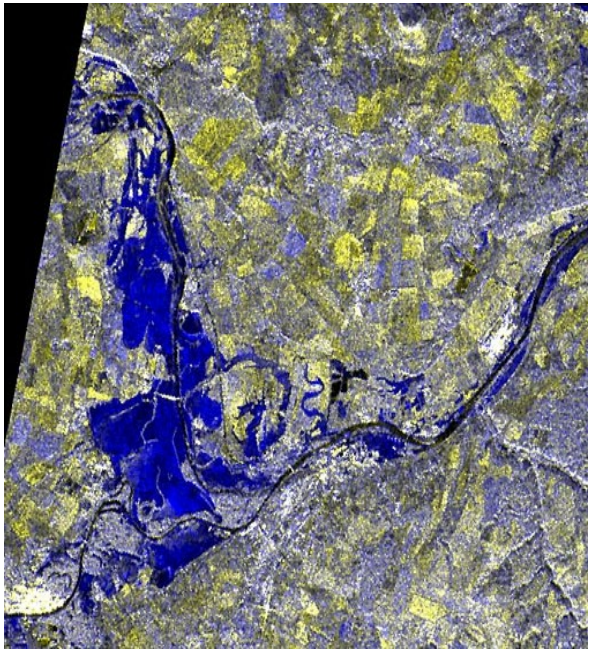
Killer application: sensitivity to surface roughness and moisture



Oil slick

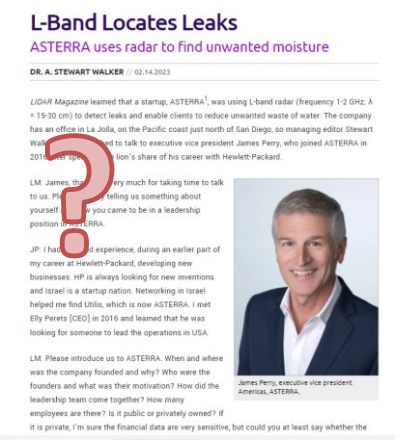


Crops & land use



Flood (color = multitemporal)





L-Band Locates Leaks

ASTERRA uses radar to find unwanted moisture

DR. A. STEWART WALKER // 02.14.2023

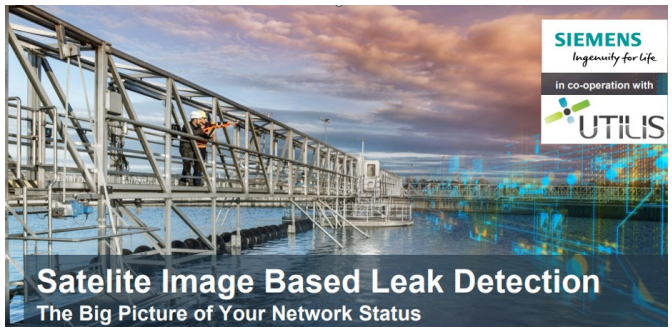
LiDAR Magazine learned that a startup, ASTERRA[1], was using L-band radar (frequency 1-2 GHz; λ = 15-30 cm) to detect leaks and enable clients to reduce unwanted waste of water. The company has an office in La Jolla, on the Pacific coast just north of San Diego, so managing editor Stewart Wall[illegible]ed to talk to executive vice president James Perry, who joined ASTERRA in 201[illegible] lion's share of his career with Hewlett-Packard.

LM: James, th[illegible]ery much for taking time to talk to us. Pl[illegible] telling us something about yourself [illegible]w you came to be in a leadership position [illegible]ERRA.

JP: I had [illegible]d experience, during an earlier part of my career at Hewlett-Packard, developing new businesses. HP is always looking for new inventions and Israel is a startup nation. Networking in Israel helped me find Utilis, which is now ASTERRA. I met Elly Perets [CEO] in 2016 and learned that he was looking for someone to lead the operations in USA.

LM: Please introduce us to ASTERRA. When and where was the company founded and why? Who were the founders and what was their motivation? How did the leadership team come together? How many employees are there? Is it public or privately owned? If it is private, I'm sure the financial data are very sensitive, but could you at least say whether the

James Perry, executive vice president Americas, ASTERRA.



Killer application: sensitivity to surface-changes

Coherent-change detection

Amplitude

Coherence (10 mins)





The first coherence map was generated by POLIMI

M. Preiss and N. J. S. Stacy, "Coherent change detection: Theoretical description and experimental results," Edinburgh, SA, Australia, Tech. Rep. DSTRO-TR-1851, 2006.

while Radar in one sentence

Human senses perceive the ‘visible’ and forms of energy that are quite different from monochromatic Radar (or Sonar, or Lidar).

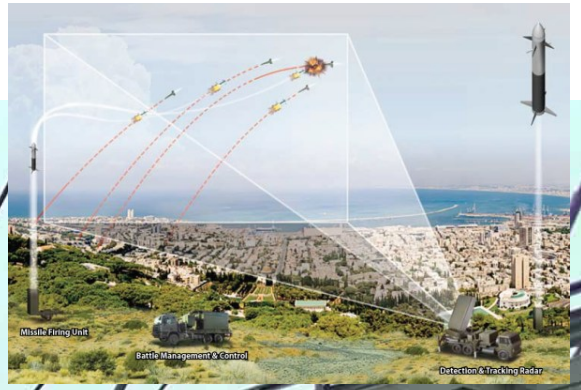
Radar SAR (or SAS) produces unfamiliar images, difficult to understand, process and interpret.

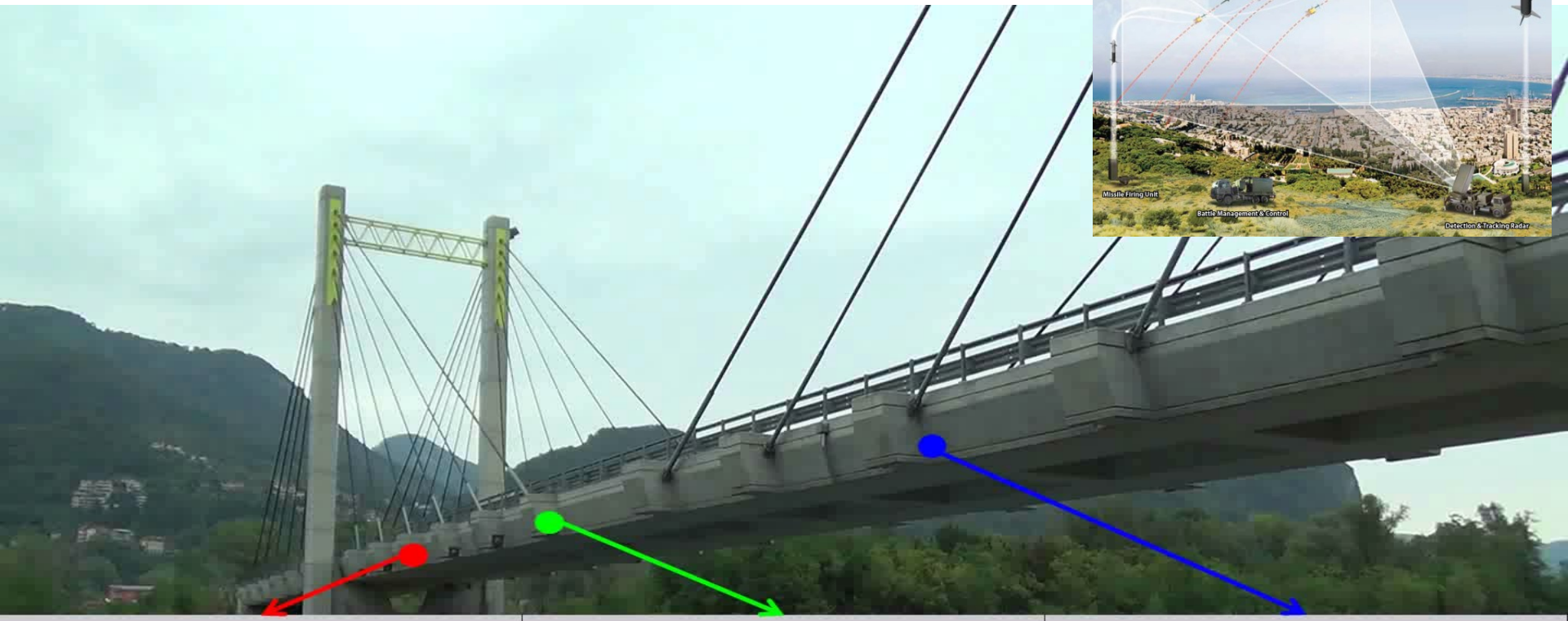
yet the technology has been exploited with success since more than 65 million year

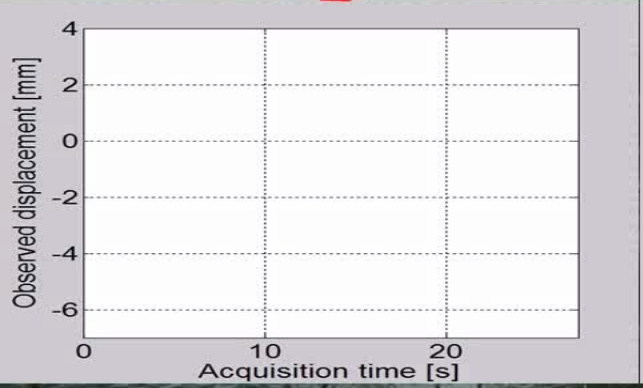


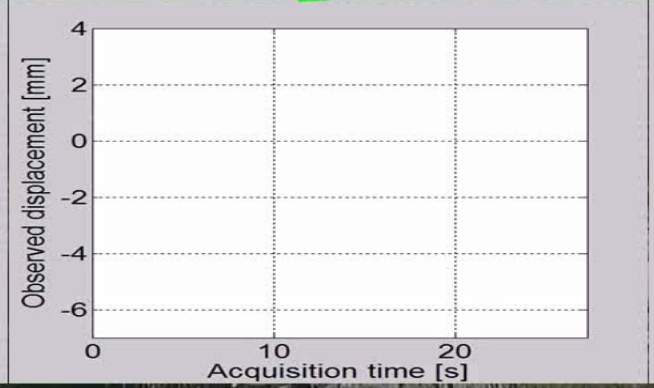
Killer application: fast and accurate Ranging

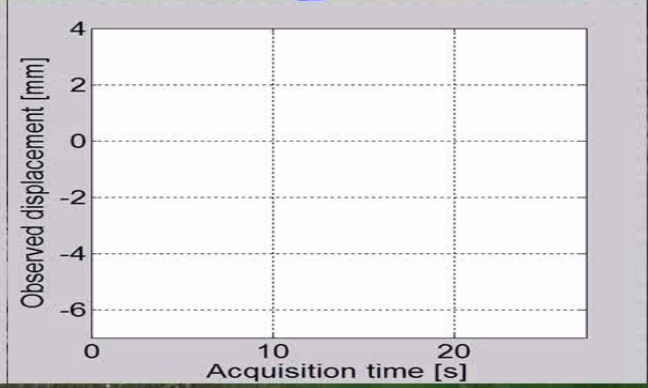
Slow-to-fast monitoring



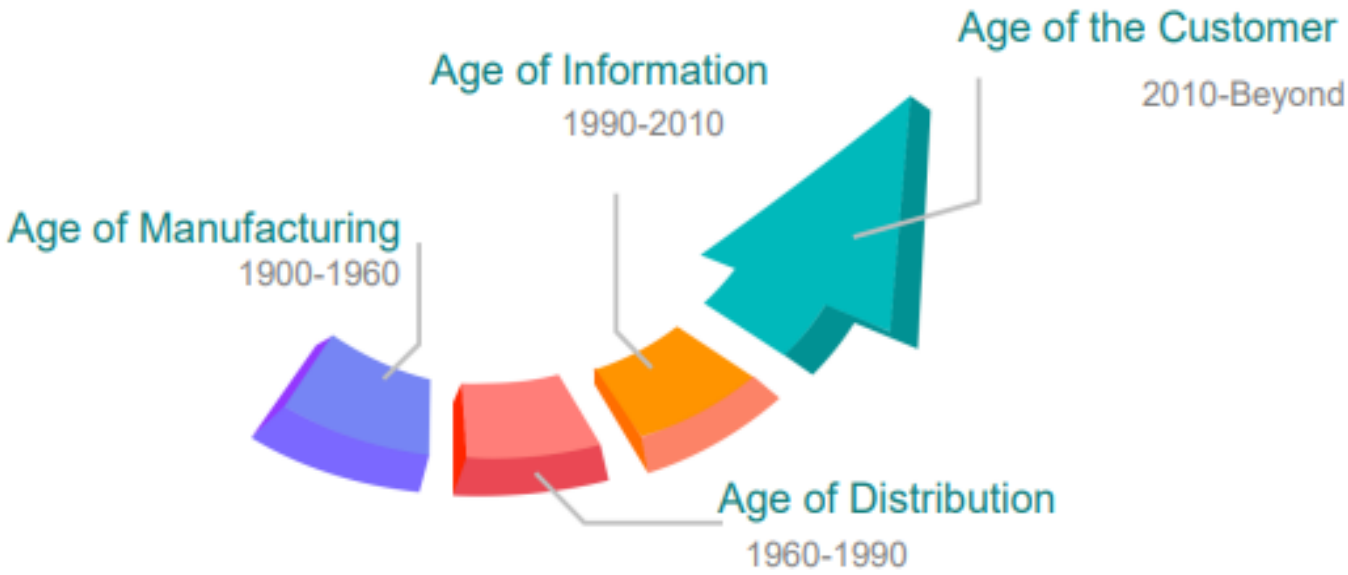








The Age of Customer is here



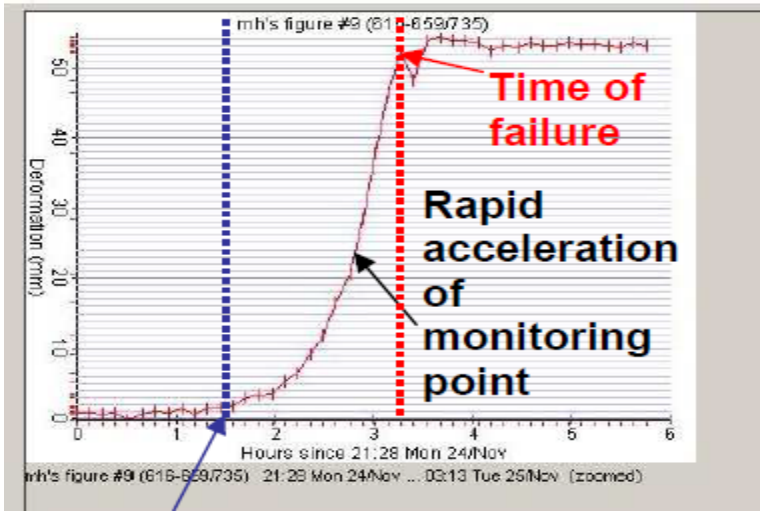
Source: « Competitive Strategy in The Age of the Customer » Forrester Report, 06/2011

Example of a Radar customized solution

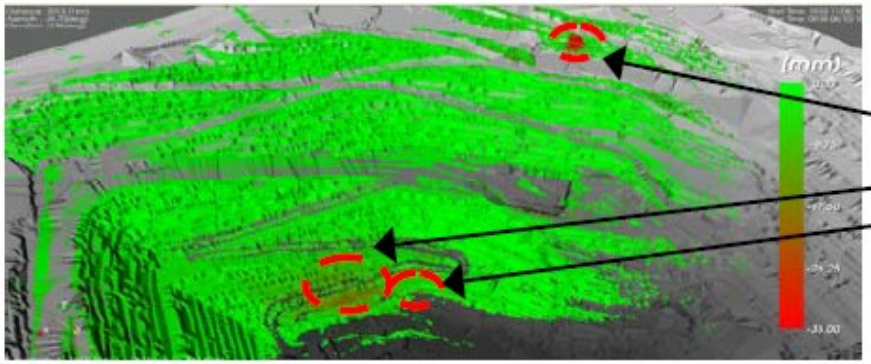
SYSTEM

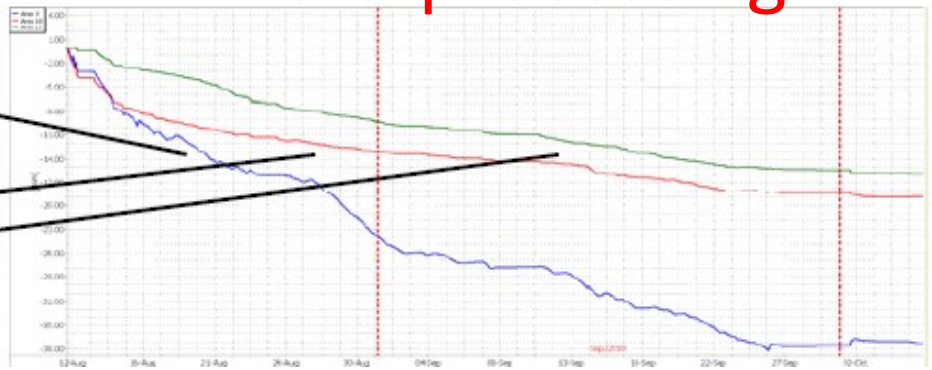


Mathematical model



Data-processing





An application

UTAH, APR. 2013. ***The biggest landslide ever in US: 165 millions tons***

Predicted by GB RADAR



http://www.ksl.com/index.php?page=1&sid=24748916&nid=460

An application

